COLLEGE OF
AGRICULTURE &
NATURAL RESOURCES
DEPARTMENT OF PLANT SCIENCE
AND LANDSCAPE ARCHITECTURE

Dear Dr. Mencuccini,

Please consider this manuscript “Early leafout extends calendar but not thermal growing seasons" as a Research Article in *Tree Physiology*. We show that longer growing seasons in calendar days lead to shorter thermal growing seasons in woody plants, with implications for how climate change may alter plant growth and carbon storage.

Most models of carbon storage assume that earlier growth will result in longer seasons and thus increase terrestrial carbon sequestration, yet recent findings have challenged this pivotal assumption (Dow *et al.*, 2022; Silvestro *et al.*, 2023), with some suggesting that woody plants may dynamically adjust the end of their growing season based on their carbon-sink capacity (Zani *et al.*, 2020; Zohner *et al.*, 2023). These observations suggest that variation in the calendar growing season (number of days) should be independent of variation in the thermal growing season (i.e., the period of favorable meteorological conditions for plant growth, Körner *et al.*, 2023). This, however, has not been tested given the paucity of start and end of season phenological events for adult trees.

Here we report on new data from a multi-species, multi-population common garden that shows longer calendar growing seasons lead to shorter (i.e., cooler) thermal growing seasons. We found that earlier leafout leads to earlier growth cessation (budset, similar results found with leaf colouring), causing longer calendar growing seasons to translate to shorter (cooler) thermal growing seasons. Because thermal conditions in the early spring are far less optimal than later-season conditions, we show that the thermal time gained by early leafout was not offset by the thermal time lost by earlier budset. Our results are based on rarely available plant-scale phenological measurements from three years across 13 woody species, all native and co-occurring in Eastern United States forests, and sampled from four populations spanning 3.5° latitude.

Our findings offer a mechanistic explanation for why multiple studies have failed to find correlations between longer calendar seasons and increased plant growth (Dow *et al.*, 2022; Silvestro *et al.*, 2023), and questions previous assertions of sink-source control in experiments (Zani *et al.*, 2020; Norby, 2021; Zohner *et al.*, 2023). At the same time, our results challenge the decades-old dogma that growth cessation is locally adapted and driven mainly by photoperiod.

We believe this work would be of broad interest to the readers of *Tree Physiology* and addresses an important debate in the field of climate change biology. The main text of this manuscript is 2449 words in length and it contains three figures. It is co-authored by C.J. Chamberlain, Deirdre Loughnan and E.M. Wolkovich and is not under consideration elsewhere. We hope that you will find it suitable for publication in *Tree Physiology*, and look forward to hearing from you.

Sincerely,

Daniel Buonaiuto

References

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